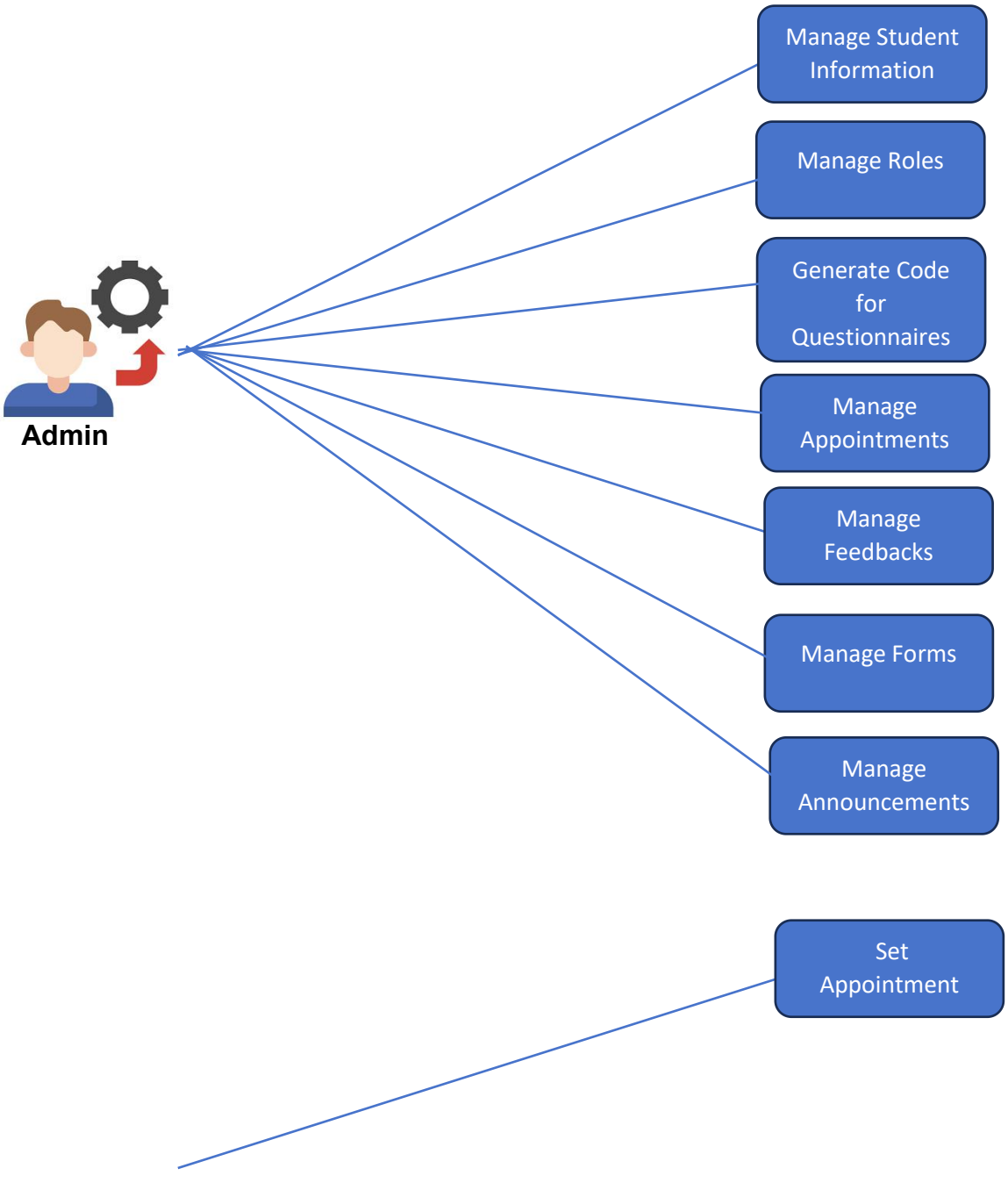
Methodology

4.1 Requirements Analysis and Requirements Documentation

In describing how the project is designed, the developers used a use-case diagram to explain how the project functions.

Use Case Diagram



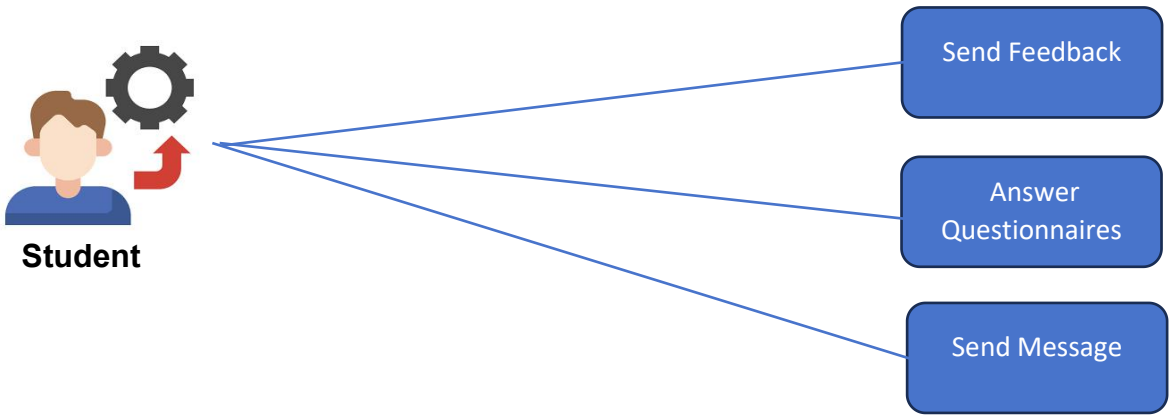


Figure 4.1.1 User Diagram

Use Case Description

Use Case Name: Manage Roles	ID:1	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: Admin registers a user then assigned to be and administrator role into the system.		
Brief Description: This use case describes how the admin registers a user then assigned to be an administrator role into the system.		
Trigger: The admin registers a user then assigned to be an administrator role into the system.		

Normal Flow of Events: 1. The admin starts the application and request to log in. 2. The admin enters username & password to the system. 3. The system navigates to dashboard. 4. The admin clicks the user's sidebar function. 5. The admin clicks the Users button. 6. The admin registers a user and assign to an administrator role. 7. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.1 Manage Roles

Use Case Name: Manage Student's Information	ID:2	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	

Scenario: The admin wants to view or update the user's information in the system.
Brief Description: This use case describes how the admin manage the student's information in the system.
Trigger: The admin wants to view or update the student's information.
Normal Flow of Events: 1. The admin starts the application and request to log in. 2. The admin enters username & password to the system. 3. The system navigates to dashboard. 4. The admin clicks the student inventory sidebar function. 5. The admin can now view and manage the student's information in the system. 6. The system ends.
Sub Flows: View user's log
Alternative/Exceptional Flows:

Table 4.1.2 Manage Student's Information.

Use Case Name: Generate Access Code for Questionnaires	ID: 3	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to generate an access code for the students to answer a particular.		
Brief Description: This use case describes how the administrator can generate an access code for the students to use for them to answer the questionnaire.		
Trigger: The Administrator can generate an access code.		
Normal Flow of Events: 1. The Administrator starts the application and log in to the system.		

2. The administrator selects what particular group or student. 3. Click the Generate Access Code function to generate an access code. 4. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.3 Generate Access Code for Questionnaires

Use Case Name: Manage Appointments	ID: 4	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to approve or cancel an appointment and create one.		
Brief Description: This use case describes how the administrator can approve or cancel an appointment and can create an appointment.		
Trigger: The Administrator can approve or cancel an appointment and create one.		
Normal Flow of Events: 1. The Administrator starts the application and log in to the system.		

5. The Administrator navigates the sidebar. 6. The Administrator clicks the Appointments tab in the sidebar. 7. Click the Approve and Cancel an appointment. 8. If administrator wants to create an appointment admin, clicks the Create Appointment button. 9. Fills up all the fields. 10. Clicks submit Appointment 11. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.4 Manage Appointments

Use Case Name: Manage Feedback	ID: 5	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: The Administrator wants to see feedbacks.		
Brief Description: This use case describes how the Administrator can see feedbacks on the services of the Guidance Office		
Trigger: The Administrator can see feedbacks.		
Normal Flow of Events: 1. The Administrator starts the application and log in to the system.		

2. The Administrator navigates the sidebar. 3. Clicks the feedback tab. 4. Feedbacks Appears. 5. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.5 Manage Feedback

Use Case Name: Manage Forms	ID: 6	Important Level: High
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: The can download forms and fills up forms.		
Brief Description: This use case describes how the Administrator can manage or download forms and fill them up.		
Trigger: The can download forms and fill them up.		
Normal Flow of Events: 1. The Administrator starts the application and log in to the system.		

2. The Administrator navigates on the sidebar. 3. Clicks the Forms/ Downloadable forms 4. Clicks Download 5. Clicks add Form 6. Clicks Submit 7. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.6 Manage Forms

Use Case Name: Manage Announcements	ID: 7	Important Level: Medium
Primary Actor: Administrator	Use Case Type: Essential, Detail	
Scenario: The Administrator can add Announcements.		
Brief Description: This use case describes how the Administrator can add Announcements.		
Trigger: The Administrator adds an Announcement.		
Normal Flow of Events: 1. The Administrator starts the application and log in to the system.		

2. The Administrator navigates on the sidebar. 3. Clicks the Announcement function. 4. Clicks Add Announcement. 5. Fills up the fields. 6. Clicks Submit. 7. The system ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.7 Manage Announcements

Use Case Name: Set Appointment	ID: 8	Important Level: Medium
Primary Actor: Student	Use Case Type: Essential, Detail	
Scenario: The student wants to set an appointment.		
Brief Description: This use case describes how the student can set an appointment.		
Trigger: The Student can sent an appointment.		
Normal Flow of Events: 1. The student starts the application.		
2. The student navigates the navbar. 3. Clicks the Set Appointment function. 4. Enters the student ID number and Name. 5. Clicks Enter. 6. Fills up the fields including the email field. 7. Clicks submit. 8. Prompts appear that the appointment has been submitted for review or approval. 9. Students can receive email stating if the appointment is approved. 10. System ends.		
Sub Flows:		

Alternative/Exceptional Flows:

Table 4.1.8 Set Appointment

Use Case Name: Send Feedback	ID: 9	Important Level: Medium
Primary Actor: Student	Use Case Type: Essential, Detail	
Scenario: The student wants to send a feedback.		
Brief Description: This use case describes how the student can send a feedback..		
Trigger: The Student can send a feedback.		
Normal Flow of Events: 1. The student starts the application.		

11. The student navigates the navbar. 12. Clicks the Feedback function. 13. Enters the student ID number and Name. 14. Clicks Enter. 15. Fills up the field. 16. Clicks submit. 17. Prompts appear that the feedback has been submitted. 18. System ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.9 Send Feedback

Use Case Name: Answer Questionnaires	ID: 10	Important Level: Medium

Primary Actor: Student	Use Case Type: Essential, Detail
Scenario: The student wants to answer a questionnaire.	
Brief Description: This use case describes how the student can answer the questionnaire.	
Trigger: The Student can answer questionnaire.	
Normal Flow of Events: 1. The student starts the application.	
2. The student navigates the navbar. 3. Clicks the Questionnaire function. 4. Enters the student ID number and access code. 5. Clicks Enter. 6. Answers the questionnaire. 7. Clicks submit. 8. Prompts appear that the questionnaire has been submitted. 9. Students can receive email stating if the appointment is approved. 10. System ends.	
Sub Flows:	
Alternative/Exceptional Flows:	

Table 4.1.10 Answer Questionnaire

Use Case Name: Send Messages	ID: 11	Important Level: Medium
Primary Actor: Administrator, Student	Use Case Type: Essential, Detail	
Scenario: The student and administrator want to send a message.		
Brief Description: This use case describes how the student and administrator can send messages.		

Trigger: The student and administrator can send a message.
Normal Flow of Events: 1. The Student and Administrator starts the application and log into the system.
2. The student and administrator navigate the sidebar. 3. Clicks the Chats function. 4. Search name of person to chat. 5. Put or create a message. 6. Clicks send. 7. System ends.
Sub Flows:
Alternative/Exceptional Flows:

Table 4.1.11 Send Messages

4.1 Design, Software and Processes

Web Application Developers want to show what to be expected in the system. It is vital or important for the developers to guide them on how they develop the system or web application for them to understand how the system works. It serves as the blue print of the system.

Output and User Interface Design

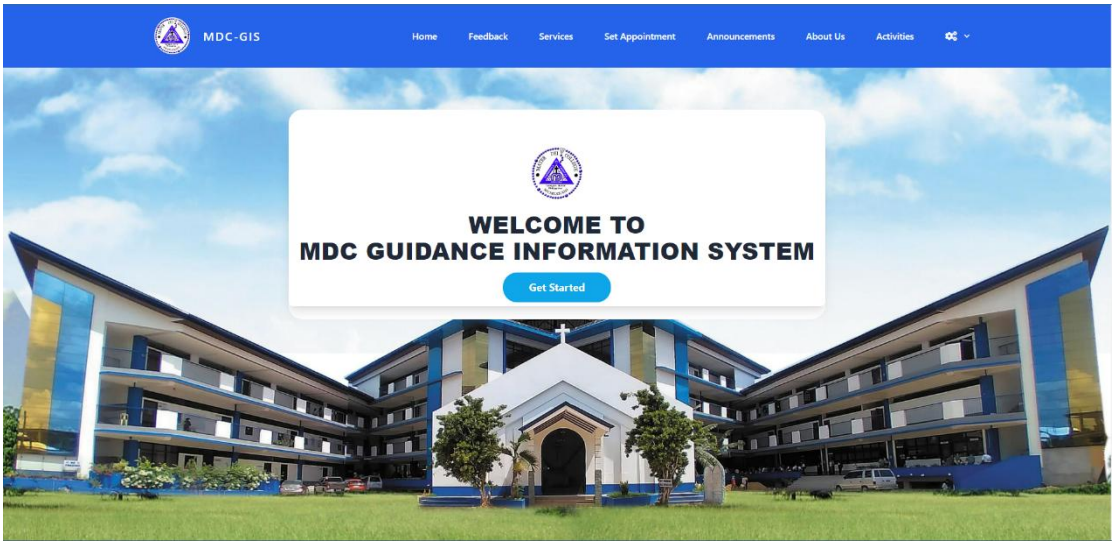


Figure 4.2.1 Home Interface

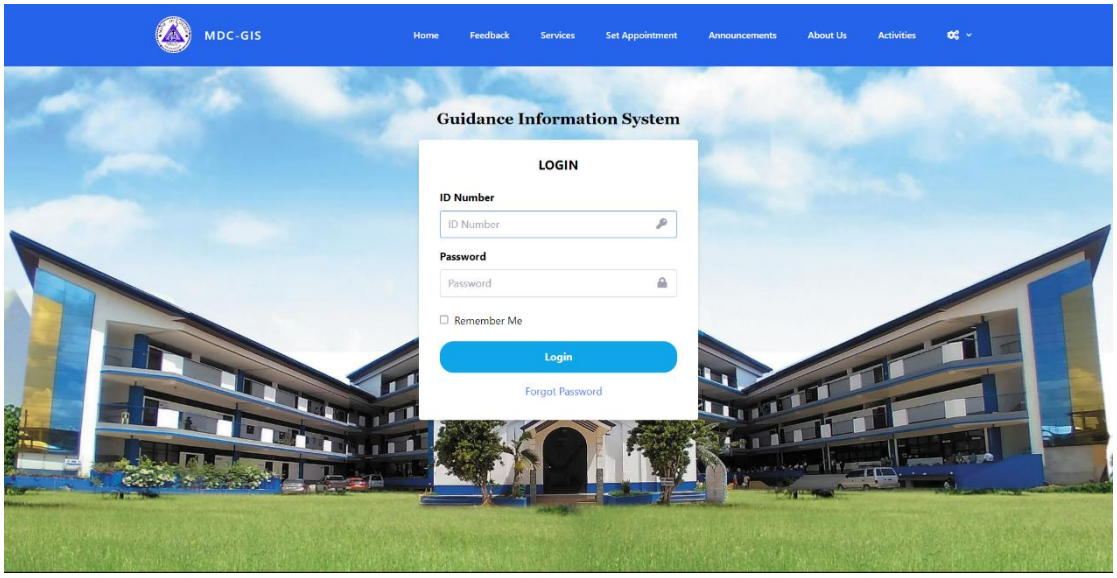


Figure 4.2.2 Log in Interface

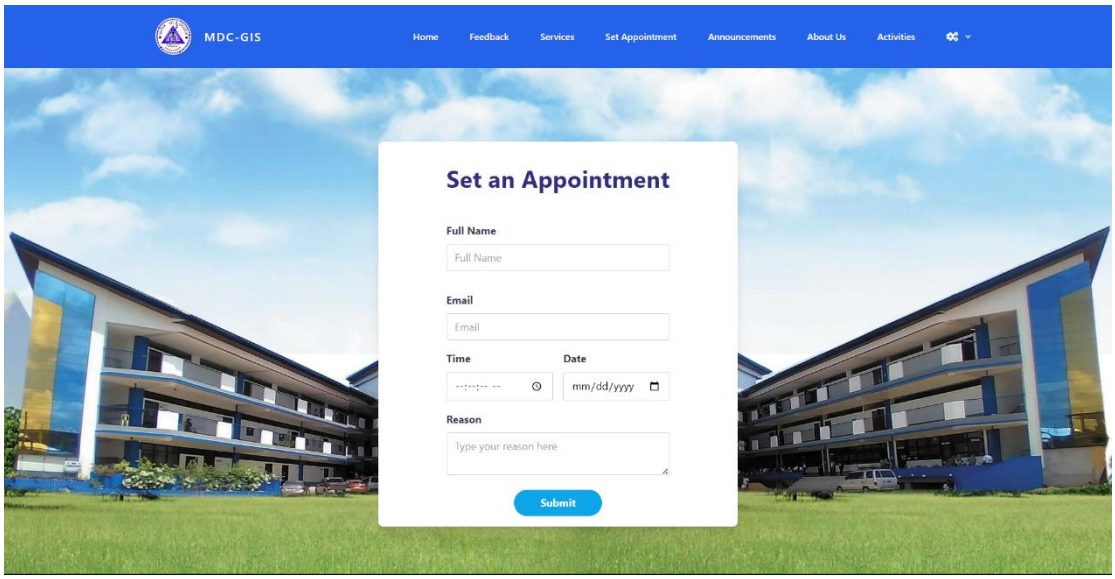


Figure 4.2.3 Log in Interface

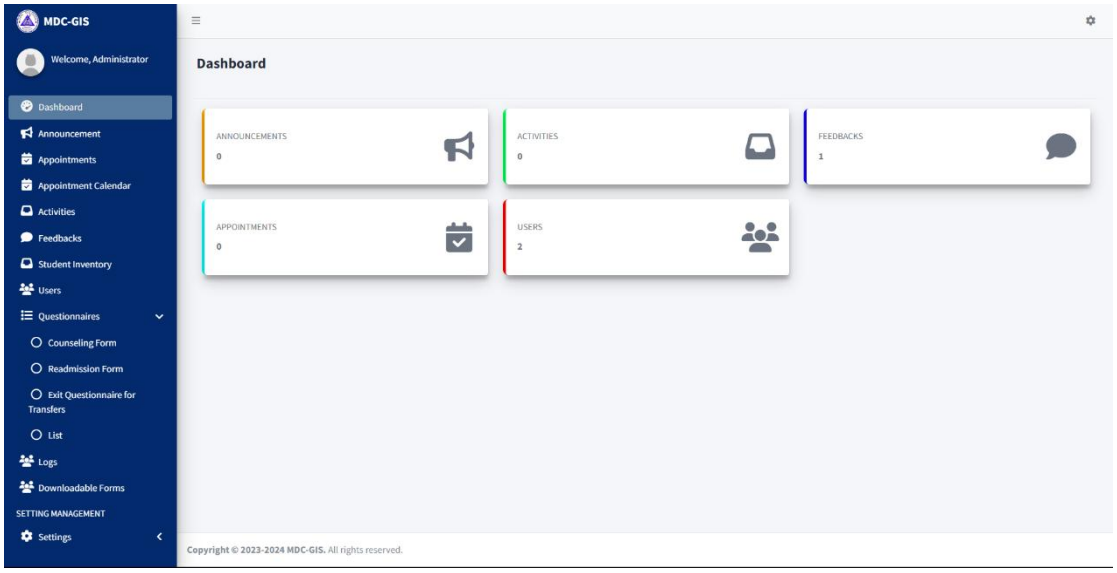


Figure 4.2.4 Admin (Dashboard) Interface

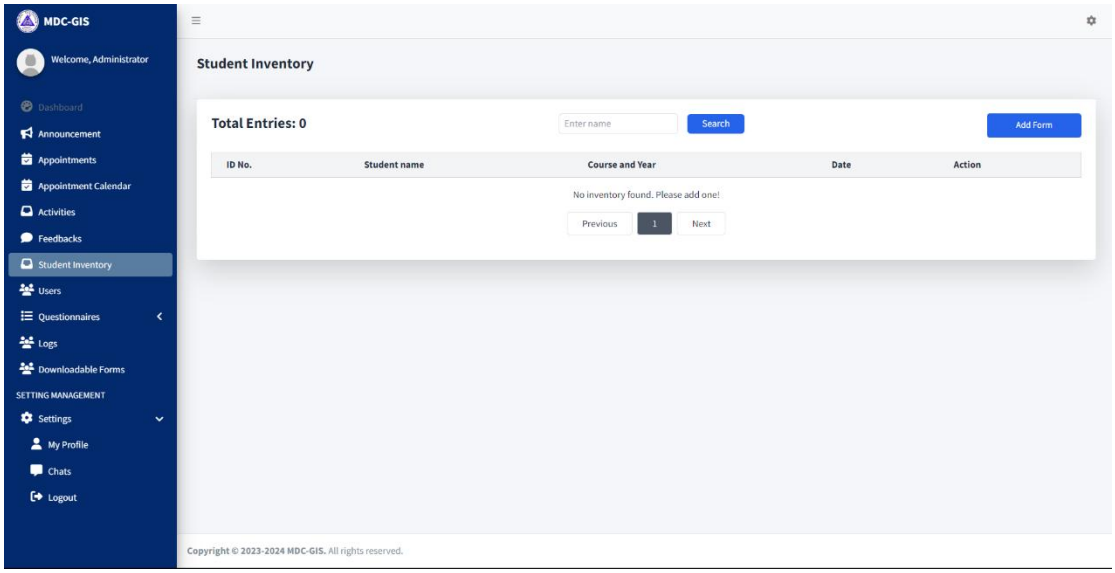


Figure 4.2.1 Student Inventory (Admin) Interface

4.3 Entity Relationship Diagram

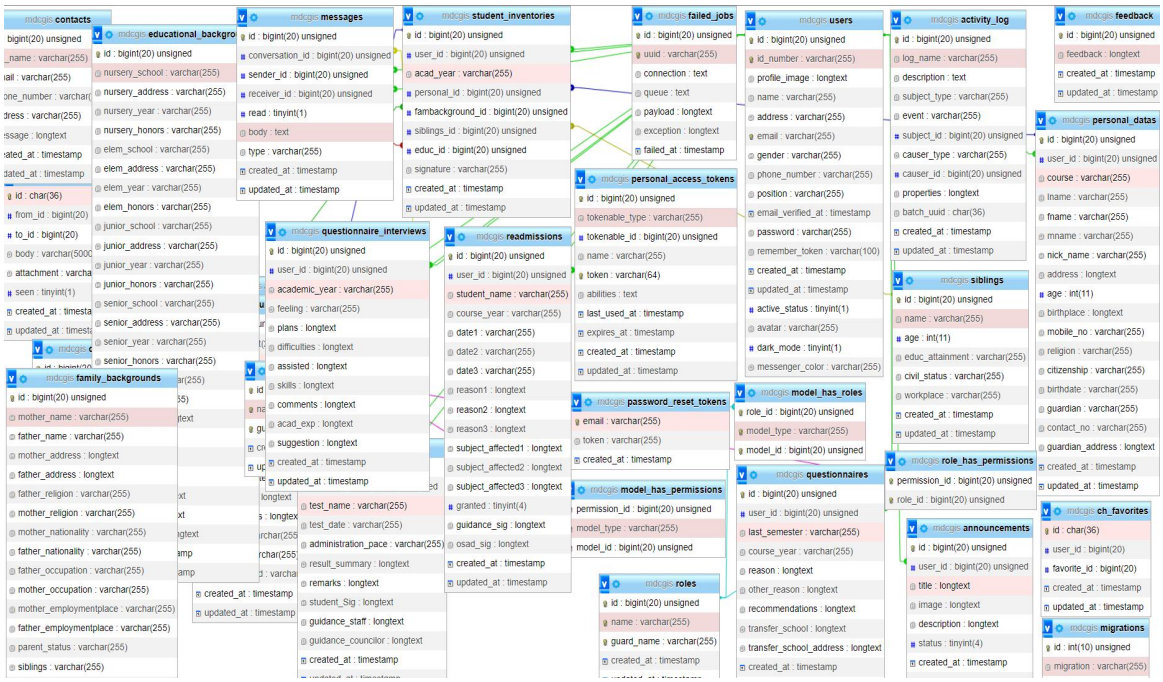


Figure 4.3.1 Home Interface

4.4 Data Dictionary

To further understand the data being saved in the database, the data dictionary will help to show the fields of the tables.

Field Name	Data Type	Length	Description
id	bigInt	20	
Id_number	varchar	255	
Profile_image	longtext		
Name	varchar	255	
address	varchar	255	
email	varchar	255	
gender	varchar	255	
Phone_number	varchar	255	
Email_verified_at	timestamp	255	
password	varchar	255	
Remember_token	varchar	100	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.1 Database Table of User

Field Name	Data Type	Length	Description
email	varchar	255	
token	varchar	255	
Created_at	Timestamp		

Table 4.4.2 Database Table of Password token

Field Name	Data Type	Length	Description
id	bigint	20	
uuid	varchar	255	
connection	text		
queue	text		
payload	longtext		
exception	longtext		
Failed_at	timestamp		

Table 4.4.3 Database Table of Failed Jobs

Field Name	Data Type	Length	Description
id	bigint	20	
Tokenable_type	varchar	255	
Tokenable_id	bigint	20	
name	varchar	255	
token	varchar	64	
abilities	text		
Last_used_at	timestamp		
Expires_at	timestamp		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.4 Database Table of Personal Access Tokens

Field Name	Data Type	Length	Description
id	bigint	20	
name	varchar	255	
Guard_name	varchar	255	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.5 Database Table of Permissions

Field Name	Data Type	Length	Description
id	bigint	20	
title	varchar	255	
remarks	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.6 Database Table of announcements

Field Name	Data Type	Length	Description
id	bigint	20	

Full_name	varchar	255	
email	varchar	255	
Phone_number	varchar	255	
address	varchar	255	
message	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.6 Database Table of Contacts

Field Name	Data Type	Length	Description
id	bigint	20	
User_id	bigint	20	
fullname	varchar	255	
date	varchar	255	
time	varchar	255	
reason	longtext		
status	tinyint	4	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.7 Database Table of Appointments

Field Name	Data Type	Length	Description
id	**bigint**	**20**	
Sender_id	**bigint**	**20**	
Reciever_id	**bigint**	**20**	
Last_time_message	**timestamp**		
Created_at	**timestamp**		
Updated_at	**timestamp**		

Table 4.4.8 Database Table of Conversation

Field Name	Data Type	Length	Description
id	bigInt	20	
Conversation_id	bigInt	20	
Sender_id	bigInt	20	
Receiver_id	bigInt	20	
read	tinyint	1	
body	text		
type	varchar	255	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.9 Database Table of Messages

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
feedback	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.10 Database Table of Feedback

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
title	longtext		
image	longtext		
description	longtext		
status	tinyint	4	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.11 Database Table of Activities

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
reason	longtext		
recommendations	longtext		
Transfer_school	longtext		
Transfer_school_address	longtext		
signature	longtext		

Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.12 Database Table of Questionnaires

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
Student_name	varchar	255	
Course_year	varchar	255	
date	varchar	255	
reason	longtext		
Subject_affected	longtext		
granted	tinyint	4	
Guidance_sig	longtext		
osad_sig	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.13 Database Table of Re admissions

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
Student_name	varchar	255	
Course_year	varchar	255	
department	varchar	255	
Contact_num	varchar	255	
Visit_nature	varchar	255	

concern	longtext		
Action_taken	longtext		
Followup_dates	longtext		
counselee	varchar	255	
counselor	varchar	255	
Session_ended	varchar	255	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.14 Database Table of Counseling Forms

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
Academic_year	varchar	255	
feeling	varchar	255	
plans	longtext		
difficulties	longtext		
assisted	longtext		
skills	longtext		
comments	longtext		
Acad_exp	longtext		
suggestion	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.15 Database Table of Questionnaire Interviews

Field Name	Data Type	Length	Description
id	bigInt	20	
User_id	bigInt	20	
course	varchar	255	
lname	varchar	255	
fname	varchar	255	
mname	varchar	255	
Nick_name	varchar	255	
address	longtext		
age	int	11	
birthplace	longtext		
Mobile_no	varchar	255	
religion	varchar	255	
citizenship	varchar	255	
birthdate	varchar	255	
guardian	varchar	255	
Contact_no	varchar	255	
Guardian_address	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.16 Database Table of Personal Data

Field Name	Data Type	Length	Description
id	bigInt	20	
Mother_name	varchar	255	
Father_name	varchar	255	
Mother_address	longtext		

Father_address	longtext		
Father_religion	varchar	255	
Mother_religion	varchar	255	
Mother_nationality	varchar	255	
Father_nationality	varchar	255	
Father_occupation	varchar	255	
Mother_occupation	varchar	255	
Mother_employmentplace	varchar	255	
Father_employmentplace	varchar	255	
Parent_status	varchar	255	
siblings	varchar	255	
Created_at	Timestamp		
Updated_at	timestamp		

Table 4.4.17 Database Table of Family Backgrounds

Field Name	Data Type	Length	Description
d	bigInt	20	
name	varchar	255	
age	int	11	
Educ_attainment	varchar	255	
Civil_status	varchar	255	
workplace	varchar	255	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.18 Database Table of Siblings

Field Name	Data Type	Length	Description

id	bigint	20	
Nursery_school	varchar	255	
Nursery_address	varchar	255	
Nursery_year	varchar	255	
Nursery_honors	varchar	255	
Elem_school	varchar	255	
Elem_address	varchar	255	
Elem_year	varchar	255	
Elem_honors	varchar	255	
Junior_school	varchar	255	
Junior_address	varchar	255	
Junior_year	varchar	255	
Junior_honors	varchar	255	
Senior_school	varchar	255	
Senior_address	varchar	255	
Senior_year	varchar	255	
Senior_honors	varchar	255	
Scholarship	varchar	255	
awards	varchar	255	
organizations	longtext		
talents	longtext		
athletics	longtext		
pasttime	longtext		
Fav_subject	longtext		
Fav_reason	longtext		
Personal_traits	longtext		
Created_at	timestamp		

Updated-at	timestamp		

Table 4.4.19 Database Table of Educational Backgrounds

Field Name	Data Type	Length	Description
id	bigint	20	
User_id	bigint	20	
Acad_year	varchar	255	
Personal_id	bigint	20	
Fambackground_id	bigint	20	
Siblings_id	bigint	20	
Educ_id	bigint	20	
signature	varchar	255	
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.19 Database Table of Student Inventories

Field Name	Data Type	Length	Description
id	bigint	20	
Inventory_id	bigint	20	
Test_name	varchar	255	
Test_date	varchar	255	
Administration_pace	varchar	255	
Result_summary	longtext		
remarks	longtext		
Student_Sig	longtext		
Guidance_staff	longtext		
Guidance_councilor	longtext		
Created_at	timestamp		
Updated_at	timestamp		

Table 4.4.19 Database Table of For Councilors

4.5 Development and Testing

Maintaining the system's or application's functionality requires constant evaluation and verification. To determine the system's efficacy, the client must first test and assess the system. A system that satisfies or surpasses the expectations of clients results in positive interactions and increased productivity.

4.6 Implementation Plan

The implementation plan's goal is to help the client come up with a method for carrying out the project or how to use is it. Comprehending the impact and significance of the offered measures in augmenting the efficacy and productivity of employees in their job is vital.

Implementation Issues and Challenges

Several complications and issues are being encountered while implementing the system. The challenges a little bit difficult and it is identified and being fixed immediately.

4.7 System Implementation & Results

A well-defined plan is the first step towards a successful implementation. This entails being aware of the actual capabilities of the system and appropriately preparing the project in question. The development tools that will be utilized in the MDC Guidance Information System must be installed before any work can begin. A well-planned and developed system will lead to a successful implementation of the system.